

Aditya Mittha

+91-8010542551 | adityamittha09@gmail.com | linkedin.com/in/adityamittha | Solapur, India

Summary

Final-year engineering student with practical experience building embedded systems around Cortex-M and ESP32 microcontrollers. Solid foundation in C programming and real-time operating systems, with a genuine interest in automation and IoT products. Comfortable debugging firmware, working close to hardware, and collaborating with cross-functional R&D teams. Eager to contribute to reliable, high-quality connected home solutions.

Skills

Programming Languages: C, Embedded C, Python, Basic Java, Verilog

Microcontrollers & Hardware: ESP32, STM32, ARM Cortex-M, Arduino, Logic Analyzer, Oscilloscope

Protocols & Communication: UART, SPI, I2C, CAN, MQTT (QoS), GPIO, Interrupts, DMA

Embedded & RTOS: FreeRTOS, Embedded Linux, Bootloader, SWD/JTAG Debugging, Sensor Networks

Cloud & Tools: Firebase, AWS IoT Core, Git, GitHub, TensorFlow, Machine Learning, Data Structures

Testing & Debugging: UART Logging, Peripheral Validation (I2C/SPI), GPIO State Checks, Unit/Integration Testing

Soft Skills: Team Collaboration, Problem Solving, Adaptability, Communication, Self-Learning

Education

Bachelor's Degree in Electronics and Telecommunication

Walchand Institute Of Technology

2027

CGPA: 9.27

Higher Schooling

D.H.B Soni College, Solapur

2023

Percentage: 83%

Work Experience

Embedded System Intern

Codec Technologies India

May 2025 – June 2025

Remote

- Delivered 2 embedded modules end-to-end by integrating microcontrollers with sensors and peripherals, meeting field-ready deployment requirements.
- Brought up UART, I²C, and GPIO interfaces; reduced board-to-board communication failures through improved initialization and error handling.
- Resolved hardware interfacing and integration defects with the team, fixing 10+ functional bugs before production release.
- Managed version control with Git/GitHub across 2 active branches, maintaining project documentation and development workflows.

Technologies: C, Embedded C, UART, SPI, I²C, GPIO, Embedded Linux, SWD Debugging, Git, GitHub

Projects

1. AQUANOVA – Smart Water Pressure Management for Equitable Water Supply

2026

- Wrote ESP32 firmware to poll pressure and flow sensors over ADC/GPIO, publishing readings via MQTT (QoS) to Firebase cloud storage for real-time pipeline monitoring.
- Built a Python analytics pipeline with pandas, numpy, and scikit-learn's Isolation Forest — detected leaks and pressure anomalies, reducing estimated maintenance response time.

Role: IoT Firmware, Embedded Systems, ML Integration

Tech Stack: ESP32, MQTT (QoS), Firebase, ADC, Pressure/Flow Sensors, Python, Telegram/SMS alerts

2. SORTIFY – AI-Powered Smart Waste Segregation System

2025

- Designed an AI-driven waste sorting system combining computer vision, weight sensing, and vibration-based sorting to classify household waste into 4 recyclable categories.
- Integrated Raspberry Pi with camera module, weight sensor, and servo motor; automated identification, measurement, and bin control using Python and C.
- Built a IoT dashboard tracking real-time segregation statistics, giving municipal teams actionable waste management data.

Role: Embedded Systems, IoT, Computer Vision Integration

Tech Stack: Raspberry Pi, Embedded C, Python, Camera Module, HX711, Servo/PWM, Firebase, IoT Cloud

3. SIGN – Real-Time Indian Traffic Sign Detection Using YOLOv11

2025

- Trained YOLOv11 Nano on 40 Indian traffic sign classes — achieved 95.9% mAP50, 95.7% precision, and 98.2% recall via transfer learning on a merged multi-source dataset.
- Built an end-to-end data pipeline: dataset merging, deduplication, balanced sampling, and weather augmentations for real-world robustness.
- Optimized inference to 15–30 FPS on CPU; deployed to Raspberry Pi via ONNX and TensorFlow Lite for edge AI applications.

Role: Machine Learning, Computer Vision, Edge AI

Tech Stack: Python, YOLOv11 Nano, PyTorch, OpenCV, NumPy, ONNX, TensorFlow Lite, Google Colab